

3/13/2025 via FDMS

Paul DeStefano, [REDACTED]

Re: AI Action Plan Dear Office of Science and Technology Policy and Faisal D'Souza, I have 15 years experience teaching undergraduate physics students, and my research is focused on how to achieve better outcomes for students through optimal learning techniques. In addition, I am a member of the PauseAI movement that opposes the development of super-human or smarter-than-human AI until we are highly confident we can ensure it serves the interest of American citizens. While I am concerned about many dimensions of AI, I am contacting you about a few key aspects of security: the risk to worker capabilities, the risk to the domestic consumer economy, and the risks to national security. As an educator, I care deeply about *how* my students learn, not merely what they learn. I've also written a lot of code, and I received my Certified Information Systems Security Professional certification in 2003. While my colleagues are actively working to create best practice for using AI to improve student learning of physics and other sciences, this work and advancements in education will not prevent AI developers from building dangerous systems that are *treated* as much more reliable than they truly are. We know that AI capabilities are over estimated by experts, professionals, and lay people. Most people struggle to use AI safely, unable to detect "hallucinations", errors, and lies presented as truth. Even these early LLMs are falsely viewed by the public as "neutral" agents with access to perfect knowledge. This conception, alone, is dangerous, even for those who never interact with AI, themselves. Our work force need to be well educated. AI may, but is not guaranteed, to improve worker capabilities and productivity. AI is NOT a tool, like the computer that brought large productivity gains to workers in many industries. AI is an agent, and workers and students interact with it in a completely different way than they do with tools. No one knows how this will turn out. While some people are excited about new technology and prefer to support the cutting edge for no other reason that it is new and exciting, our economy relies on stable and smooth execution. Right now, the economy is becoming paralyzed by the anticipated disruption of AI. Businesses need to know that, whatever they imagine is possible through AI, the policy of the U.S. toward AI will deliver robust competition, safety, security, and fairness. And consumers need the same! The federal government should not purchase AI tools that have not been reviewed for safety and security by experts. To that end, it should work to establish an independent body that establishes guidelines and requirements for the industry. Finally, AI poses a serious threat to our national security. Maintaining global dominance in artificial intelligence technology will be difficult, and we should hedge our bets by earnestly participating in the global effort to govern AI technology. Experts have repeatedly expressed genuine concern over the threat to all human life that we face with AI. Personally, I am *not* afraid of this eventuality; I don't think this is likely. But, this outcome is so catastrophic that we must invest deeply in preventing it, no matter how unlikely. This is an inherent risk for this technology. We have to take it seriously. Does that investment mean greatly increasing the current cost of AI, and therefore forgoing future benefits of AI advancements? Yes. But that doesn't change the calculation. Recommendations Many good ideas have been proposed to ensure AI is developed responsibly. Here are some of my favorites: - We need an international body of experts to track and verify the AI capabilities, worldwide. Much like nuclear weapons, which are not available to just anyone with the knowledge, powerful AI remains far beyond consumer access and control. The U.S. can benefit from being a strong member of such an effort. - We need legislation that

places the full burden of all types of cost on developers. - Extremely strong whistleblower protections. Sincerely,